Case 18: Temporal convergence

1 Verification purpose

This isolated-process case verifies temporal convergence of ELMFIRE's level-set propagation for a manufactured circular front. Four time grids vary the whole-second step while mesh and physical fields remain fixed. The case can expose timestep scaling errors, inconsistent final-time handling, or a non-convergent temporal update. The question is whether all four simulation conditions complete, the fitted radius-error order is at least 0.8, and the finest-step radius error is at most 0.5 m.

2 Mathematical formulation and numerical implementation

The governing equation is $\partial_t \phi + R_0 |\nabla \phi| = 0$, with constant $R_0 = 0.0116332$ m s^{-1}. All simulation conditions share the same 1024-by-1024, 256 m raster grid. For a 360 s simulation, the configured time-grid counts 45, 90, 180, and 360 correspond to whole-second $\Delta t = \{8, 4, 2, 1\}$ s.

3 Derivation of expected behavior

The exact final radius is $r_\star = 20 + (0.0116332)(360) = 24.187952$ m. Define $E_{\Delta t} = |r_A - r_\star|$. A consistent time integration has $E_{\Delta t} \approx C(\Delta t)^p$; a least-squares slope of $\log E_{\Delta t}$ versus $\log \Delta t$ estimates the observed temporal order p.

The time-step simulation conditions share the manufactured reference

$$r_{\text{ref}} = r_0 + R_s T = 20 + (0.0116332)(360) = 24.187952 \text{ m}.$$

For $\Delta t_i \in \{8, 4, 2, 1\}$ s, the corresponding exact step counts are $360/\Delta t_i = \{45, 90, 180, 360\}$. From each final raster,

$$r_i = \sqrt{\frac{N_{b,i}(0.5 \text{ m})^2}{\pi}}, \qquad E_i = |r_i - 24.187952|.$$

The reported temporal order is the least-squares log–log slope

$$p_t = \frac{\sum_i (\ln \Delta t_i - \overline{\ln \Delta t})(\ln E_i - \overline{\ln E})}{\sum_i (\ln \Delta t_i - \overline{\ln \Delta t})^2}.$$

The substituted decision tests are $p_t \geq 0.8$ and $E_{\Delta t=1} = |r_1 - 24.187952| \leq 0.5$ m.

4 Verification metrics and acceptance criteria

Observed p uses all four simulation conditions and is required to be at least 0.8, allowing front-raster measurement effects while demanding near-first-order convergence. Finest-step error must be no more than 0.5 m, preventing a formally favorable but inaccurate sequence. Equivalent radius is obtained from the largest closed, linearly interpolated $\phi = 0$ contour, avoiding binary cell-count bias. The last matching level-set field raster for every time grid is required. All simulation conditions and both metrics must pass; missing data is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

All EPSG:32610 simulation conditions use a 256 m square domain, 0.25 m cells, uniform inputs, flat terrain, zero wind, centered 20 m signed-distance ignition, and duration 360 s. Only the time-grid control varies. The predicted circular-front radius approaches 24.187952 m as Δt decreases.

Parameter	Configured value	Units	Role
Domain/grid	256 m square; 0.25	m/cell	Fixed spatial discretization
Time steps	8, 4, 2, 1	s	Temporal refinement sequence
Stop time	360	s	Common exact comparison time
Initial front	centered signed-distance radius 20	m	Common exact geometry
Rate	0.0116332	m/s	Manufactured normal speed

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

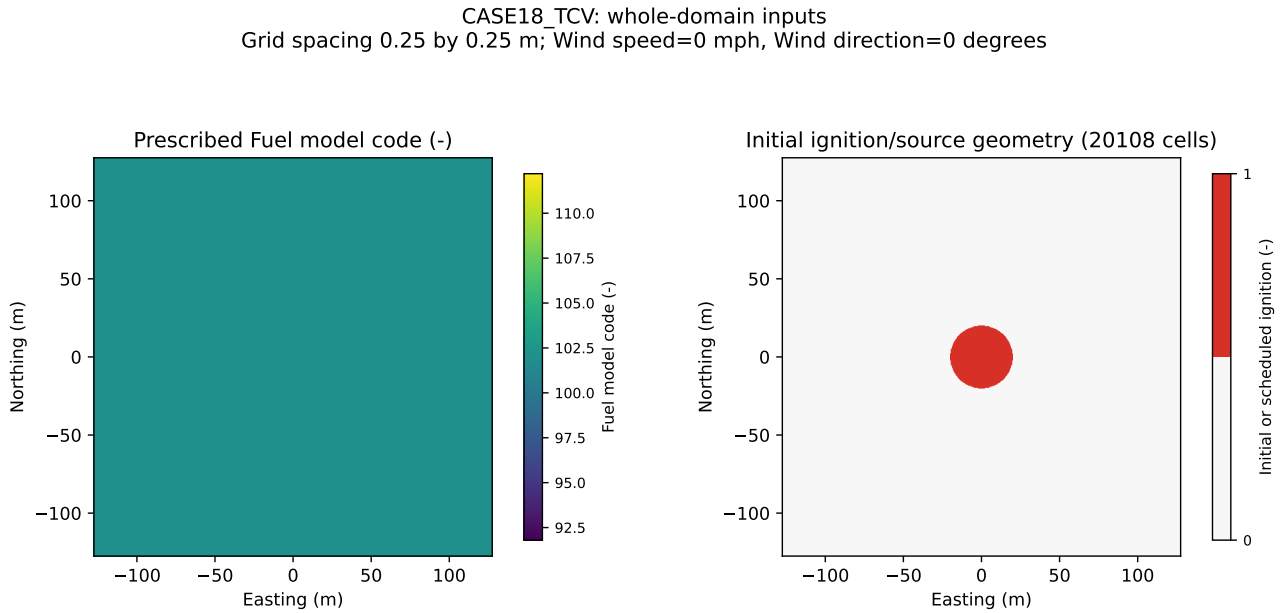


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-specific output rasters are read. Current status is **FAIL**; 4 simulation conditions and 2 decision metrics are recorded.

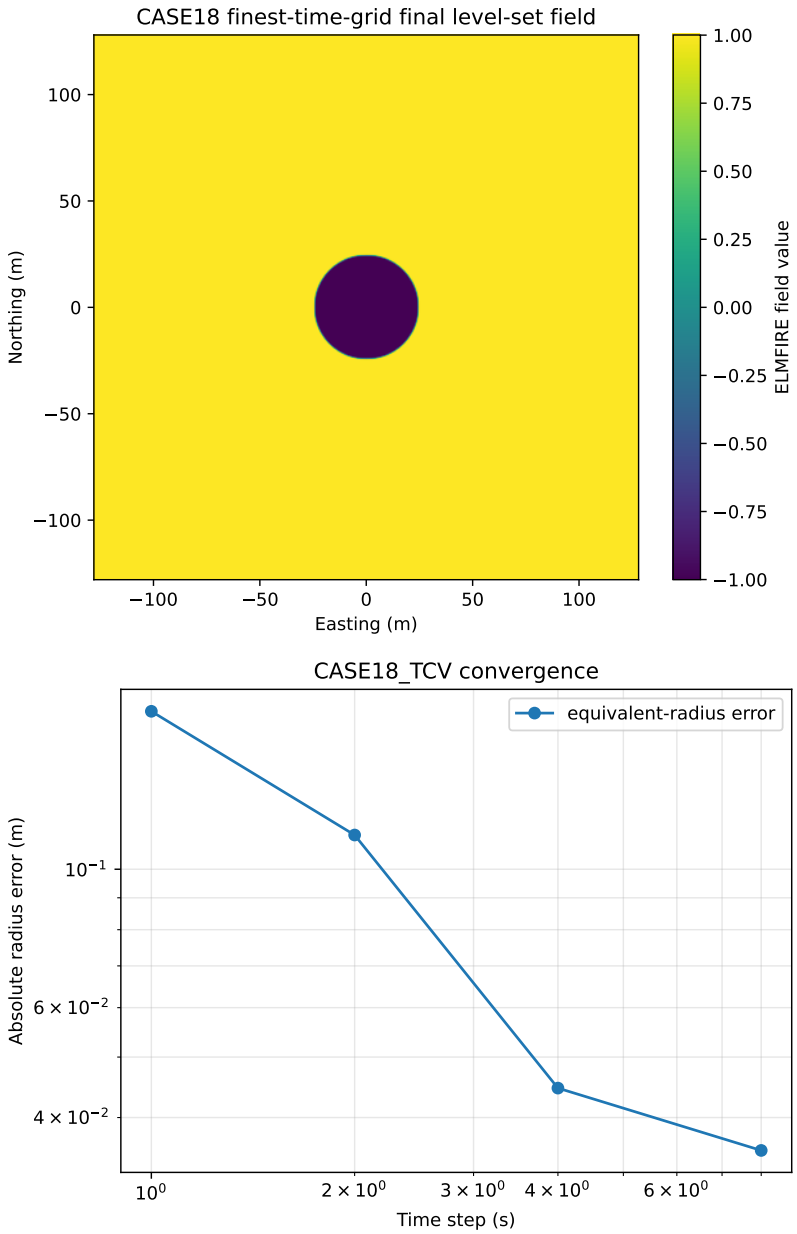


Figure 2: Finest-step whole-domain signed level-set result and radius-error convergence across all four completed time grids. The spatial field is dimensionless; radius errors are in metres, and the generated table supplies both acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
observed temporal order	0.8 ($\geq$ threshold)	-0.836072 -	FAIL
finest-time-grid radius error	0 ($\leq$ 0.5)	0.179096 m	PASS

The authoritative overall decision is **FAIL**. The generated table is sourced directly from the case-specific calculated metrics and is not recomputed in this report.

8 Technical interpretation

Low order can indicate timestep application, output-time selection, or pre-asymptotic spatial error; a small finest error alone does not establish convergence. Missing simulation conditions invalidate the fitted order and are treated as insufficient evidence.

9 Scope and limitations

This case verifies temporal behavior for one smooth deterministic front at a fixed fine mesh. It does not separate all spatial error, test adaptive timestep logic under complex forcing, or provide physical validation.